

Kyle Main

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Senior Data Engineer — Large-Scale Data Pipelines & Cloud Analytics

Data engineer with 12 years of experience designing and operating the data pipelines, canonical datasets, and data-quality systems that power analytics and downstream decision-making at scale. Built data ingestion and pipelining for 220+ unique data sources from the ground up as an early startup hire, including a Common Information Model (data dictionary) standardizing field names and types across every pipeline, plus a homegrown Apache Beam program on GCP Dataflow for historical/cold-storage data retrieval. Deep hands-on experience with PySpark, GCP Dataproc, and BigQuery for large-scale distributed data processing and analysis. Comfortable working across many stakeholder teams to translate business and product needs into reliable, fault-tolerant data infrastructure.

CORE SKILLS

Data Engineering & Pipelines: Data ingestion/pipelining for 220+ sources; Common Information Model / canonical dataset & data-dictionary design and standardization; data-quality monitoring and alerting; connector/collector health monitoring and troubleshooting; ETL parsing/normalization (50+ Logstash filters, homegrown parsing tooling)

Distributed & Cloud Data Processing: PySpark / SparkSQL, GCP Dataproc compute clusters, Apache Beam on GCP Dataflow (historical/cold-storage data retrieval), BigQuery, Zeppelin notebooks; working exposure to Kafka and Flink in a production streaming environment

Programming & Analysis: Python, SQL, Git; Pandas, NumPy, SciPy, scikit-learn; time-series and clustering analysis at scale

Cloud Platforms: GCP (Dataproc, Dataflow, BigQuery, serverless/event-driven data enrichment), AWS, Azure

Data Governance & Cross-Team Delivery: Data security/integrity/compliance monitoring; schema and access governance across heterogeneous platforms; sustained delivery working across engineering, data science, and business stakeholder teams

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new data ingestion platform from the ground up for a federal client — three distinct high-volume telemetry sources (CrowdStrike, Suricata, Zeek) pipelined into a central Elasticsearch analytics environment, including data transforms, a custom analytics/detection layer built on top of the pipeline output, data-quality monitoring and alerting content, and custom dashboards (DOE/NNSA Security Data Integration project, completed).
- Currently create and manage analytics content that turns raw event data into decision-ready outputs for a SOC's case and incident workflows — the same data-to-insight pipeline discipline this role's canonical-dataset and metrics work requires (Treasury SOC / TSSOC, current project).
- Supported data ingestion and data-quality efforts within an Elasticsearch/Splunk environment for a federal continuous-monitoring program (CISA CDM at DOE, completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and engineering team processing and analyzing massive-scale customer data on cloud-based big-data tooling, building statistical, time-series, and ML-based analytical content on top of that data pipeline.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Owned data engineering for a next-gen cloud analytics platform at massive scale as a very early startup hire — built ingestion/pipelining for 220+ unique data sources, 50+ Logstash parsing/normalization filters, and a Common Information Model (data dictionary) standardizing field names and types across every pipeline; also assisted building "Loggify," a homegrown log-parsing/filtering tool that replaced Logstash, plus ongoing connector/collector health monitoring and troubleshooting.
- Built a homegrown Apache Beam program run on GCP Dataflow to fetch large volumes of historical cold-storage data for customers — real hands-on distributed batch-processing pipeline ownership, not just streaming ingestion.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters using Zeppelin notebooks and a home-grown reusable analysis toolkit — Spark jobs loading data from cloud storage buckets, PySpark/SparkSQL for analysis, supporting statistical, behavioral, and ML-based (device clustering) analytical content.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale datasets to build custom detection models — DNS-based malware detection/mitigation and anomalous-behavior discovery across the network.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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August 3, 2026

Hiring Team
OpenAI

I'd like to bring my data engineering background to the Applied team's work building the pipelines and canonical datasets that power OpenAI's analyses, safety systems, and product growth.

As one of the earliest engineers at Cysiv, a cybersecurity startup later acquired by Trend Micro and then Forescout, I built the data engineering foundation for our entire analytics platform from scratch: ingestion and pipelining for 220+ unique data sources, 50+ parsing/normalization filters, and a Common Information Model — a canonical data dictionary standardizing field names and types across every pipeline we ran. That last piece maps directly onto this role's charge to develop canonical datasets that track key metrics across a fast-growing product. I also built a homegrown Apache Beam program run on GCP Dataflow to retrieve large volumes of historical cold-storage data for customers, and ran exploratory data analysis at scale on GCP Dataproc clusters using PySpark and SparkSQL — the same distributed-processing muscle this role calls for with Spark.

Throughout my career, I've worked cross-functionally with infrastructure, data science, and product stakeholders to turn raw, high-volume event data into reliable, fault-tolerant systems that downstream teams could trust for decision-making — the same collaborative, data-integrity-first approach this role requires working across Infrastructure, Data Science, Product, Marketing, Finance, and Research. I'm drawn to the chance to apply that background to data that directly powers safety systems and the researchers building OpenAI's models.

I'd welcome the opportunity to talk through how this experience applies to building OpenAI's next data pipelines.

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